

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

THESIS ASSIGNMENT SHEET OF BACHELOR PROGRAM

Semester 2025.2

Student's information

Full name:	Duong Phuong Thao	Student ID:	20226001
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Supervisor's information

Full name:	Pham Quang Dung		
Thesis carried out at:	School of Information and Communications Technology		
Time duration for thesis:	From	24/2/2026	to 15/6/2026

1. Thesis's title:

Building a software for distribution management

2. Thesis's field of interest:

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|---|-----------|---------------------|
| - | Option 1: | Enterprise Software |
| - | Option 2: | |
| - | Option 3: | |
| - If the field of interest is not listed in the available options, the supervisor may recommend a suitable topic: | | |

3. Objectives**3.1. Knowledge acquired:**

- Software development process for supply chain management systems: surveying, analysis, design, implementation and testing.
- Supply chain management concepts: procurement, sales, inventory, logistics and accounting.
- Full-stack web application development: Backend API, Frontend SPA, Database.

3.2. Technologies acquired:

- Java, Spring Boot framework ecosystem (Spring Security, Spring Data JPA).
- React.js for building user interfaces (SPA).
- PostgreSQL, JWT, Docker, Railway Cloud Deployment.

3.3. Skills developed:

- Communication and requirements analysis skills.
- Time management and independent working.
- Technical report writing and presentation skills.

3.4. Products expected:

- A complete supply chain management system supporting: Procurement, Sales, Inventory, Logistics, Accounting.
- Responsive web interface with role-based access control (RBAC).
- Dashboard for revenue, inventory and debt overview.

3.5. Practical issues solved:

- Growing demand for integrated supply chain management for SMEs.
- Manual inventory processes leading to errors and lack of control.
- Need for real-time data solutions for business decision-making.

4. Tentative work packages and plan:

Note: Required credits for a bachelor-graduation thesis are 6(0-0-12-12), i.e. 12 hours per week for 17 weeks.

Work package 1: Literature survey,	from Week	1	to Week	2
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Details:

- Define objectives, target users, and existing problems in supply chain management.
- Survey existing SCM software: SAP, Odoo, KiotViet.
- Identify related technologies and documentation.

Work package 2: Study/Research on related technologies,	from Week	3	to Week	4
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Details:

- Java, Spring Boot, Spring Security, Spring Data JPA, React.js, Vite, PostgreSQL, Docker, Railway, JWT, Postman, GitHub.

Work package 3: Analysis and design,				from Week	5	to Week	7
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from Week	5	to Week	7
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[illegible]

- Detailed use-case analysis and user role definition.
- Business flow analysis for each module.
- System architecture design, database schema design, UI wireframes.

Work package 4: Program development,			from Week	8	to Week	15
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from Week	8	to Week	15
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Details:								
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- Develop modules: Product, Customer, Supplier, Warehouse, PO, GR, SO, GI, Invoice.
- Build responsive React UI, RESTful APIs, JWT auth with RBAC.
- Develop Accounting module and Dashboard. Deploy to Cloud (Railway).

Work package 5: Test and evaluation.				from Week	16	to Week	17
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from Week	16	to Week	17
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[illegible]

- Unit testing APIs, UI testing.
- Integration testing: PO → GR → SO → GI → Invoice → Payment.
- Security testing (RBAC enforcement). Collect user feedback and finalize.

5. Commitment of the student who has received the assignment					
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I commit to complete the assignment as planned above.						
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Hanoi, Date 20/4/2026

Student

(Signature and full name)

Duong Phuong Thao

6. Supervisor's confirmation of the student assignment												
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Hanoi, Date 20/4/2026

Supervisor

(Signature and full name)

Pham Quang Dung